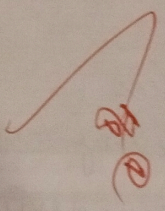
Command prompt/Router CLI:

Router>enable
Router #configure terminal.
Router (config)#ip route 192.168.2.0 255.255.255.0
Router (config)#end
Router #show ip route.

Sample output:

5 192.168.20/24 [1/0] via 10.0.0.2



Exp: 11 Routing at the Network layer

Date:- 13/9/24 (a) Static Routing Protocol configuration using Cisco packet tracer

Aim:-
To configure static routing b/w different networks using cisco packet tracer and verify packet.

Algorithm:-
1. Open Cisco packet tracer and place two routers, switches and PCs
2. Connect the PCs to their respective switches and router.
3. Assign IP address and subnet mask to the PCs and router interface.
4. Configure the IP address of the router interfaces and enable them.
5. Configure the IP addresses of the inter-router links.
6. Add static router on each router using the route command.
7. Verify the routing table [Router# show ip route]
8. Use ping to test communication b/w PCs on different networks.
9. Observe packet transmission in simulation mode

Rip routing

Router >enable
Router # configure terminal.
Router (config) # router rip
Router (config) # version 2
Router (config-router) # network 192.168.1.0
Router (config-router) # network 192.168.2.0
Router (config-router) # end.

Sample output

R 192.168.2.0/24 [120/1] via 10.0.0.2

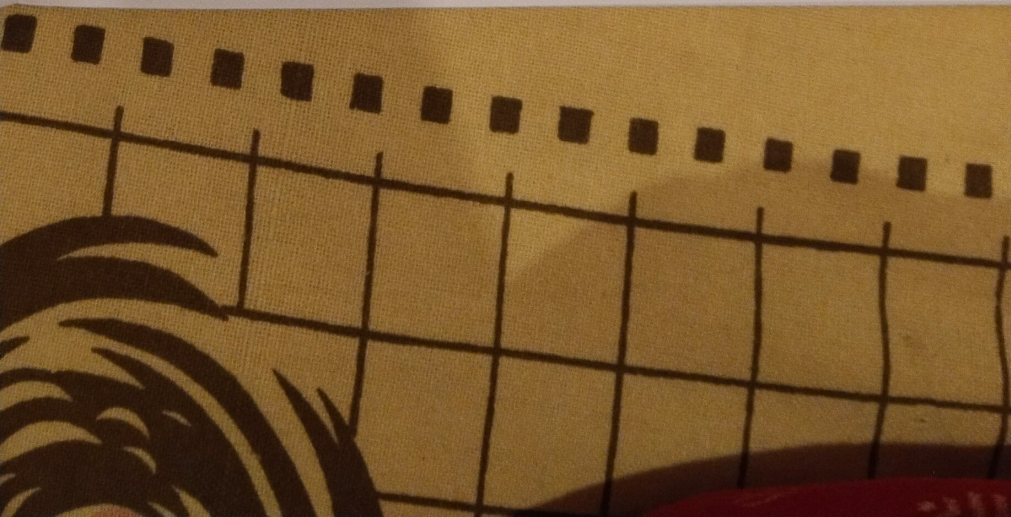
(b) RIP configuration using Cisco packet tracer.

Aim:-

To configure the Routing Information protocol (RIP) and verify dynamic routing between different networks using Cisco packet tracer.

Algorithm:-

1. Open Cisco packet tracer and create a topology with multiple routers and networks.
2. Connect the routers and end devices using suitable cables.
3. Assign IP addresses and subnet masks to all devices and router interface.
4. Enable the router interfaces using the no shutdown command.
5. Enter RIP configuration mode on each router.
6. Enable RIPv2 and advertise the directly connected networks.
7. Configure the appropriate network statements on every routers.
8. Verify the routing table using show ip route.
9. Confirm that routes learned through RIP appear with the code R.
10. Test connectivity b/w pcs in different networks using ping.



Result:-

RIP was configured successfully. The routers exchanged routing information dynamically, learned remote network, routes and forwarded packets b/w the networks.